

LumiCity Run Guide

Overview

LumiCity is a Streamlit application for smart street light monitoring, prediction, camera-based traffic response, and light intensity analysis. This project folder also includes a Jupyter notebook used for the model-building and analysis workflow.

Files to copy to another device

Copy the full project folder so these files stay together: `app.py`, `credentials.json`, `kerala_smart_lights.db`, `intensity_model.pkl`, `intensity_scaler.pkl`, `yolov8n.pt`, `requirements.txt`, `requirement.txt`, and `LumiCity_Step_by_Step_Analysis.ipynb` if you want the notebook too.

System requirements

Windows is the safest target because the app uses `cv2.CAP_DSHOW` for camera access. Install Python 3.10 to 3.13 and make sure pip is available. A webcam is only needed for the Camera Surveillance tab.

Environment setup

1. Open a terminal inside the project folder.
2. Create a virtual environment: `python -m venv .venv`
3. Activate it on PowerShell: `.venv\Scripts\Activate.ps1`
4. Upgrade pip: `python -m pip install --upgrade pip`
5. Install dependencies: `pip install -r requirements.txt`

You can also use `requirement.txt` because it contains the same package list.

Run the application

Start the app with: `streamlit run app.py`

After Streamlit starts, open the local URL shown in the terminal, usually `http://localhost:8501`.

What the app depends on at runtime

The app expects the SQLite database, model pickle files, YOLO weights file, and `credentials.json` to be present in the same folder as `app.py`. If any of these are missing, some features will fail or show empty data.

Internet access

Live weather and NASA VIIRS retrieval need internet access. If those services are unavailable, the app may fall back to defaults or show stale data depending on the workflow.

Notebook usage

If you want to open `LumiCity_Step_by_Step_Analysis.ipynb`, run: `jupyter notebook` or open the notebook in VS Code after installing the same requirements. The notebook includes

training and data-preparation steps that produced the saved model artifacts.

Common issues

If Streamlit is not recognized, use: `python -m streamlit run app.py`

If the camera tab cannot access the webcam, close other apps using the camera and rerun the app.

If YOLO downloads or model loading fail, confirm `yolov8n.pt` is present in the project folder.

If package installation fails, retry after upgrading pip and verify the active interpreter is the virtual environment.

Recommended quick check

After startup, verify the main dashboard opens, the map renders, analytics charts appear, and image upload works. Camera access is optional but should be tested on the target machine if that feature matters.